

LAVNEET HORA

707 Avenue T, Lubbock, TX 79401

☎ 832-418-0086

✉ 31lavneet@gmail.com

🌐 [linkedin.com/in/lavneethora](https://www.linkedin.com/in/lavneethora)

🐙 github.com/lavneethora

Education

Texas Tech University

Expected Graduation: May 2027

Bachelor of Science in Computer Science (Honors), Minor in Math and Bioengineering

GPA: 3.97

- Coursework: Data Structures & Algorithms, Software Engineering, Computer Architecture, Quantum Computing
- Awards: 4x President's Honor List, Dean's List, IEEE Computer Society Career Catalyst Scholar
- Activities: Honors College FYE Mentor, Engineering LYFE Mentor, Learning Assistant

Experience

UTHealth Houston, BIG-TCR Program

June 2026 – August 2026

Undergraduate Research Assistant (Incoming)

Houston, TX

- Selected for competitive Big-TCR program to conduct computational analysis of cancer genomics data using bioinformatics pipelines and statistical methods under faculty mentorship.

Texas Tech University, Dept. of Electrical and Computer Engineering

May 2025 – October 2025

Undergraduate Research Assistant

Lubbock, TX

- Identified 5 key behavioral indicators for anxiety prediction by analyzing movement patterns across 100+ patient datasets using statistical correlation analysis.
- Reduced preprocessing time by 50% by implementing automated MATLAB pipelines for DFA/MF-DFA analysis and Hurst exponent visualization.
- Standardized trend analysis methodology across 100+ daily movement profiles, improving reproducibility and reducing analysis variance by 30%.

iMocha

July 2025 – August 2025

GenAI Intern

Remote

- Reduced QA triage time by 75% (12→3 min/report) by building Gemini API classification pipeline that automated issue categorization for 500+ monthly coding assessments.
- Decreased resolution cycles by 40% by engineering Python ETL workflows that transformed 1,000+ candidate feedback entries into structured QA reports.

Projects

Quantum Chess | Python, Qiskit, Quantum Computing

April 2026

- Implemented superposition and entanglement logic using Qiskit quantum circuits, enabling pieces to occupy multiple board positions with probabilistic measurement outcomes.
- Designed game flow system including turn management, move validation, quantum state collapse handling, and win condition detection.
- Built the UI system including turn timers, game controls, board visualization, and rule validation for quantum-specific moves (split, merge, entangle).

Football Kick Analyzer | Rubik Pi 3, Python, YOLOv8, OpenCV, MediaPipe

November 2025

- Built and labeled a custom dataset of 3,000+ football images across varied lighting/angles (published on Kaggle), then trained YOLOv8 model achieving 20-35ms inference latency on embedded hardware (Rubik Pi 3).
- Engineered trajectory reconstruction pipeline using Kalman filtering and spline interpolation to smooth noisy ball detections across motion blur conditions.
- Presented system demo to Qualcomm engineers and leadership team during IEEE Career Catalyst program visit.

AI Student Assistant Chatbot | Python, React, DeepSeek, Model Tuning

March 2025

- Won the AMD Ryzen AI Track at HackCU12 for building a full-stack RAG chatbot that answered student queries using the UC Boulder handbook as a knowledge base.
- Reduced model latency by 40% through quantization and fine-tuning of DeepSeek while maintaining answer accuracy above 85%.
- Implemented document parsing and prompt engineering pipeline in 24-hour hackathon sprint.

Technical Skills

Languages: Python, Java, C++, C, SQL, MATLAB, JavaScript, R

Frameworks & Libraries: React, PyTorch, OpenCV, TensorFlow, Next.js, Node.js, FastAPI, Tailwind CSS, Pandas, NumPy, Qiskit, Selenium

Tools & Platforms: Git, Docker, Firebase, Supabase, Jupyter, VS Code, Google Cloud, Linux

Leadership / Extracurricular

HackWesTX

January 2025 – September 2025

Hacker Experience Lead

Texas Tech University

- Coordinated participant experience for 200+ attendees at Texas Tech's annual hackathon.

Google Developers Group

August 2024 – Present

Officer

Texas Tech University

- Coordinated workshop logistics and facilitated collaborations with other campus tech clubs.